

Multiple Choice (10 marks)

1. The lunar maria are:
 - (a) ancient heavily cratered highlands
 - (b) dark lavas inside volcanic calderas
 - (c) dark lavas filling older impact basins
 - (d) the bright regions on the Moon
2. Why are Cepheid stars relevant for astronomers?
 - (a) To measure interstellar mass.
 - (b) To measure galactic distances.
 - (c) To measure galactic energy-density.
 - (d) To measure interstellar density.
3. The sky is blue because
 - (a) the Sun mainly emits blue light.
 - (b) the atmosphere absorbs mostly blue light.
 - (c) molecules scatter red light more effectively than blue light.
 - (d) molecules scatter blue light more effectively than red light.
4. The terrestrial planet cores contain mostly metal because
 - (a) the entire planets are made mostly of metal.
 - (b) metals condensed first in the solar nebula and the rocks then accreted around them.
 - (c) metals sank to the center during a time when the interiors were molten throughout.
 - (d) radioactivity created metals in the core from the decay of uranium.
5. Which one of these constellations is not located along the Milky Way in the sky?
 - (a) Perseus
 - (b) Cygnus
 - (c) Scorpius
 - (d) Leo

True / False (10 marks)

Answer True or False

6. True or False: Knowing both the actual brightness and apparent brightness of an object allows one to estimate its distance from the observer.
7. True or False: The ratio of solar radiation flux on Mercury's surface at perihelion compared to aphelion is 9:4.
8. True or False: Differentiation is primarily caused by spontaneous emission from radioactive atoms within the planet's interior.
9. True or False: The visible part of the electromagnetic spectrum ranges from approximately 380 to 740 nanometers.
10. True or False: The period of rotation is the same for every point on a solid disk regardless of its distance from the axis of rotation.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of the Laniakea supercluster within the context of the Milky Way's structure and its implications for our understanding of cosmic evolution.
12. Discuss the definition of a parsec in astronomy, including its significance and how it relates to the measurement of distance using astronomical units and angular measurements.

End of Paper